Design and Vibration Analysis of a Small-Satellite Solar Panel for the Falcon 9 Rideshare Frequency Requirement

Maxat Bekmurzayev[1]

A launch vehicle shakes its payload strongly during ascent, and a payload whose natural frequency is too low can go into resonance with this excitation. For this reason the SpaceX Rideshare Payload User's Guide does not allow a payload to have natural frequencies below 40 Hz. This report describes the sizing of a simple deployable solar panel of a small satellite against this requirement. The panel was modeled as a cantilever with a tip mass. A beam finite element code was written in MATLAB and checked against the exact cantilever solution. With this code, solid aluminum, solid carbon-fiber and aluminum honeycomb sandwich panels were compared. A 6 mm solid aluminum panel was then modeled in ANSYS Mechanical with solid elements. The ANSYS first natural frequency is 46.7 Hz, which is 2.1% above the beam model and 16.7% above the requirement. A harmonic base excitation analysis with 2% damping gave a tip acceleration of 34.9 times the input and a root bending stress of 32.3 MPa per 1 g of input. Both values agree with the MATLAB model within 1%. The study also shows that a sandwich panel meets the same requirement with about 87% less mass than the solid aluminum panel.

Nomenclature

A	= cross-sectional area, m^2
a_b	= base acceleration amplitude, m/s^2
b, t, L	= panel width, thickness and length, m
E	= Young's modulus, Pa
f_i	= i-th natural frequency, Hz
I	= second moment of area, m^4
$\mathbf{K}$, $\mathbf{M}$	= stiffness and mass matrices
m_{tip}	= tip mass, kg
Q	= quality factor (amplification at resonance), $Q = 1/(2\zeta)$
$\mathbf{r}$	= rigid-body influence vector
Γ_j	= modal participation factor of mode j
ζ	= modal damping ratio
ρ	= density, kg/m^3
ϕ_j	= mass-normalized mode shape of mode j
ω, Ω	= natural and excitation circular frequency, rad/s

I. Introduction

During launch, a satellite is exposed to low-frequency sine vibration, random vibration, acoustic noise and shocks [8]. The low-frequency part comes mainly from engine thrust oscillations and the interaction between the rocket structure and the propulsion system. If a natural frequency of the satellite is close to the frequencies where the rocket itself vibrates strongly, the two structures can couple and the response of the satellite can grow by an order of magnitude or more. To avoid this, every launch provider sets a minimum allowed natural frequency for its customers. According to Section 4.1.1 of the SpaceX Rideshare Payload User's Guide, a payload must not have elastic natural frequencies below 40 Hz, and its quality factor Q should be between 10 and 50 [1].

For a small satellite, the main body is usually stiff. The parts that are most likely to violate the requirement are long and thin appendages such as solar panels, antennas and booms. In this work a deployable solar panel was selected because it is the most common appendage and because its simple shape allows a hand calculation. This makes it possible to check a commercial finite element result against an independent model. This kind of check is also done in real structural analysis.

The objectives of the work were:

1. to size a solar panel so that its first natural frequency is above 40 Hz with a design margin;

[1]Undergraduate student, L. N. Gumilyov Eurasian National University, Astana, Kazakhstan.

2. to compare solid aluminum, solid composite and honeycomb sandwich concepts by mass;
3. to verify the results with three independent methods: an analytical formula, a self-written MATLAB finite element code and an ANSYS solid model;
4. to estimate the panel response and stress under sine base excitation.

II. Problem Definition

A. Panel Model and Assumptions

The panel is a rectangular plate with length $L = 300$ mm and width $b = 100$ mm. One short edge is attached to the satellite body through a hinge bracket, and the free end carries a small concentrated mass of $m_{tip} = 50$ g, which represents a hold-down fitting and a sun sensor. The model is shown in Fig. 1.

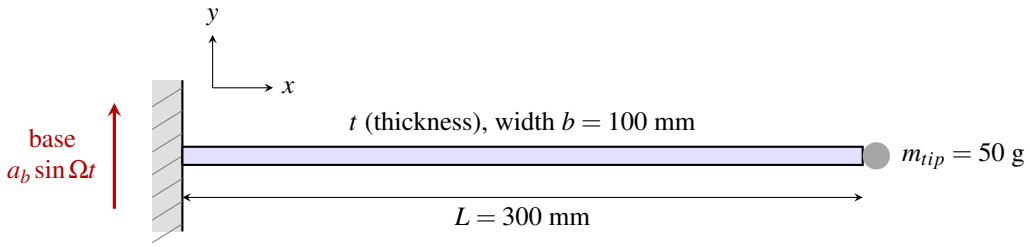


Fig. 1 Simplified panel model: cantilever plate clamped at the hinge, with a tip mass and base excitation in the thickness direction.

The following assumptions were made:

- The hinge is rigid (clamped boundary). A real hinge is flexible, so this assumption gives a frequency that is too high. This is the reason for the design margin below.
- The tip mass is a point mass without rotary inertia.
- Material behavior is linear elastic and damping is viscous with a constant modal damping ratio $\zeta = 0.02$ ($Q = 25$), which is inside the range given in the user's guide [1].
- In reality the panel is folded and held at several points during launch. The cantilever case is the softest configuration and therefore conservative.

B. Requirement and Design Target

The requirement is $f_1 \geq 40$ Hz [1]. Since the finite element models are idealized (rigid hinge, ideal material), a 10% margin was added, so the design target is $f_1 \geq 44$ Hz. Two materials were considered; their properties are listed in Table 1.

Table 1 Material properties used in the analysis

Material	E, GPa	ρ, kg/m^3	ν	Strength, MPa
Aluminum 6061-T6 [2]	68.9	2700	0.33	276 (yield)
Quasi-isotropic CFRP (typical)	50.0	1550	0.30	≈400

The CFRP properties are typical values for a quasi-isotropic carbon/epoxy laminate and depend strongly on the layup. They are used only for the concept comparison.

III. Theory

A. Natural Frequencies of a Cantilever

For a uniform Euler–Bernoulli cantilever without tip mass, the natural frequencies are [3, 4]

$$\omega_i = (\beta_i L)^2 \sqrt{\frac{EI}{\rho A L^4}}, \qquad \beta_1 L = 1.8751,\ \ \beta_2 L = 4.6941,\ \ \beta_3 L = 7.8548 \tag{1}$$

where $I = bt^3/12$ and $A = bt$. With a tip mass, a good estimate of the first frequency is obtained by the Rayleigh method, where 33/140 of the beam mass m_b is added to the tip mass [3]:

$$f_1 \approx \frac{1}{2\pi}\sqrt{\frac{3EI}{L^3\,(m_{tip} + 0.2357\,m_b)}} \tag{2}$$

B. Beam Finite Element Model

To get more than one mode and to compute the forced response, the cantilever was divided into n two-node Euler–Bernoulli beam elements with two degrees of freedom per node (deflection w and rotation θ). The element stiffness and consistent mass matrices are [5]

$$\mathbf{k}_e = \frac{EI}{L_e^3}\begin{bmatrix} 12 & 6L_e & -12 & 6L_e \\ 6L_e & 4L_e^2 & -6L_e & 2L_e^2 \\ -12 & -6L_e & 12 & -6L_e \\ 6L_e & 2L_e^2 & -6L_e & 4L_e^2 \end{bmatrix}, \qquad \mathbf{m}_e = \frac{\rho A L_e}{420}\begin{bmatrix} 156 & 22L_e & 54 & -13L_e \\ 22L_e & 4L_e^2 & 13L_e & -3L_e^2 \\ 54 & 13L_e & 156 & -22L_e \\ -13L_e & -3L_e^2 & -22L_e & 4L_e^2 \end{bmatrix} \tag{3}$$

After assembly, the tip mass is added to the deflection degree of freedom of the last node, and the two degrees of freedom of the clamped node are removed. The natural frequencies and mode shapes follow from the generalized eigenvalue problem

$$\left(\mathbf{K} - \omega^2 \mathbf{M}\right)\phi = \mathbf{0} \tag{4}$$

C. Response to Base Excitation

When the clamp moves with acceleration $a_b \sin \Omega t$, the motion $\mathbf{u}$ relative to the base satisfies

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = -\mathbf{M}\mathbf{r}\,a_b \sin \Omega t \tag{5}$$

where $\mathbf{r}$ has ones at the deflection degrees of freedom and zeros at the rotations. Using mass-normalized modes ϕ_j and modal damping, each mode responds independently with the complex amplitude [4, 6]

$$q_j = \frac{-\Gamma_j\, a_b}{\omega_j^2 - \Omega^2 + 2i\zeta\,\omega_j \Omega}, \qquad \Gamma_j = \phi_j^T \mathbf{M}\mathbf{r} \tag{6}$$

The value Γ_j^2 is the effective modal mass, i.e. the part of the mass that moves with mode j when the base is shaken. The absolute tip acceleration is $a_b - \Omega^2 u_{tip}$, and the bending stress at the root is $\sigma = E\,\kappa t/2$, where κ is the curvature at the clamp.

D. Sandwich Panel

For a sandwich panel with two face sheets of thickness t_f and a core of thickness h_c, the bending stiffness is carried almost only by the faces. With the distance between face mid-planes $d = h_c + t_f$, the bending stiffness is [7]

$$EI \approx \frac{E_f\, b\, t_f\, d^2}{2} \tag{7}$$

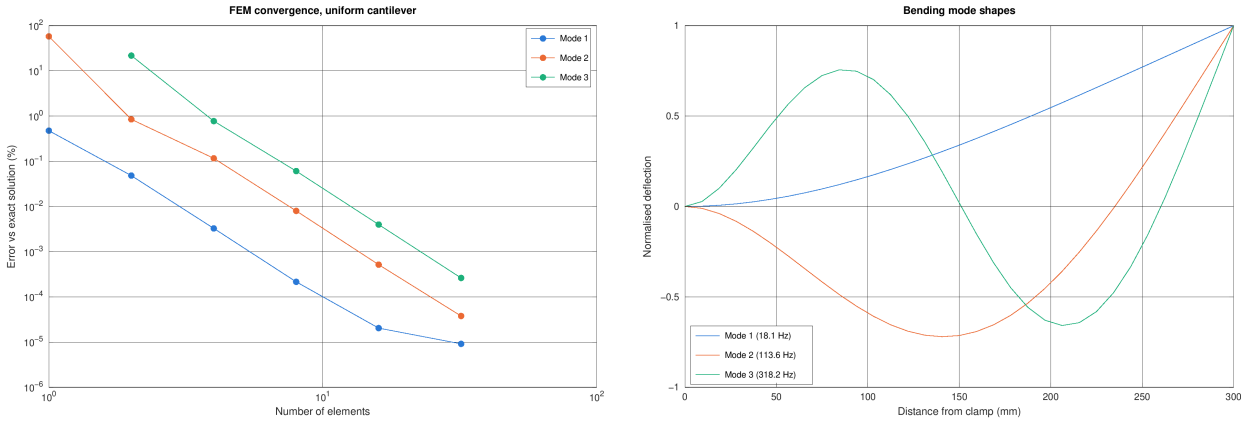
Core shear deformation was neglected, which makes the predicted frequency slightly too high.

IV. MATLAB Model and Verification

The equations above were programmed in MATLAB. Before using the code for design, it was tested on a bare aluminum cantilever ($t = 2$ mm, no tip mass), for which Eq. (1) gives the exact answer: $f_1 = 18.13$ Hz, $f_2 = 113.64$ Hz and $f_3 = 318.21$ Hz. Table 2 and Fig. 2a show that the error decreases quickly with the number of elements. With 8 elements the first frequency is already within 0.001% of the exact value. All later calculations used 40 elements.

Table 2 Convergence of the MATLAB beam model (bare cantilever, $t = 2$ mm)

Elements	f_1, Hz	Error, %	f_2, Hz	Error, %	f_3, Hz	Error, %
2	18.143	0.048	114.61	0.85	387.63	21.8
4	18.135	0.003	113.78	0.12	320.67	0.77
8	18.134	<0.001	113.65	0.008	318.40	0.061
16	18.134	<0.001	113.65	<0.001	318.22	0.004
Exact	18.134	–	113.64	–	318.21	–



(a) Error versus number of elements

(b) First three bending mode shapes

Fig. 2 Verification of the MATLAB beam model on a uniform cantilever.

V. Design Study

A. Solid Panels

The first natural frequency was computed for thicknesses from 1 to 6 mm, using commercially available sheet thicknesses. Table 3 shows the aluminum results together with the hand estimate from Eq. (2). The two methods agree within 1%, so both can be trusted. The bending stiffness grows with t^3, while the panel mass grows only with t, so f_1 grows a little faster than linearly with t. For a solid plate of a given material, the only way to raise the frequency is to make it thicker, and each extra millimeter of aluminum adds 81 g. The thinnest aluminum sheet that meets the 44 Hz target is 6 mm thick, with $f_1 = 45.7$ Hz and a mass of 486 g (Fig. 3a).

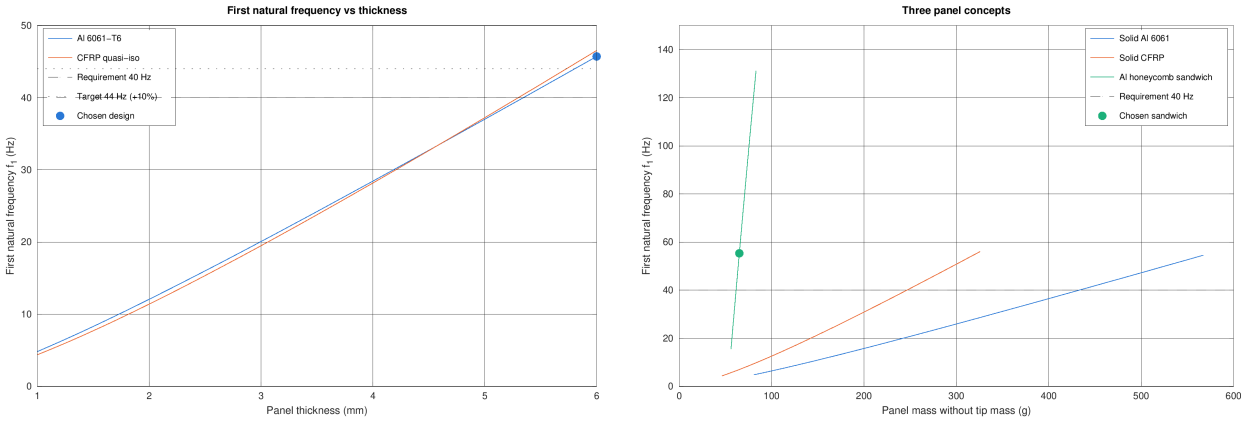
Table 3 First natural frequency of the solid aluminum panel with 50 g tip mass

t, mm	f_1 FEM, Hz	f_1 Eq. (2), Hz	Panel mass, g	$f_1 \geq 44$ Hz?
2.0	12.07	12.11	162	no
3.0	20.08	20.17	243	no
4.0	28.46	28.61	324	no
5.0	37.03	37.27	405	no
6.0	**45.73**	46.06	486	**yes**

A solid CFRP panel behaves in a similar way. It needs the same 6 mm thickness ($f_1 = 46.6$ Hz) but weighs 279 g, because carbon fiber has a higher stiffness-to-density ratio than aluminum. Even so, both solid panels are heavy for a small satellite.

B. Honeycomb Sandwich Panel

Real solar panels are not solid plates but sandwich panels. In a second design iteration, a panel with two 0.3 mm aluminum face sheets and an aluminum honeycomb core (density 50 kg/m^3) was analyzed. An 8 mm core gives $f_1 = 55.3$ Hz at a panel mass of only 65 g, including an assumed adhesive mass of 0.15 kg/m^2. Compared with the 6 mm solid aluminum panel, this is an 87% mass reduction while the frequency is 21% higher. Fig. 3b shows why. For the sandwich, a small increase in mass gives a large increase in frequency, because the stiffness grows with the square of the core thickness, and the core itself is very light.



(a) f_1 versus thickness, solid panels

(b) f_1 versus panel mass, three concepts

Fig. 3 Design study results. The dashed line is the 40 Hz requirement.

For the ANSYS verification the 6 mm solid aluminum panel was chosen, because it is isotropic, easy to model and can be compared directly with the beam model. The sandwich panel is the recommended design, and its verification in ANSYS is left for future work.

VI. ANSYS Finite Element Model

A. Model Setup

The panel geometry ($300 \times 100 \times 6$ mm) was created in SolidWorks and imported into ANSYS Mechanical 2026 R1 (Student). A custom material with the aluminum properties of Table 1 was defined, because the default ANSYS aluminum alloy has slightly different values. The panel was meshed with hexahedral solid elements (Fig. 4). The hinge end face was fixed (Fixed Support), and the 50 g tip mass was applied as a Point Mass on the free end face with deformable behavior, so that it acts at the center of that face.

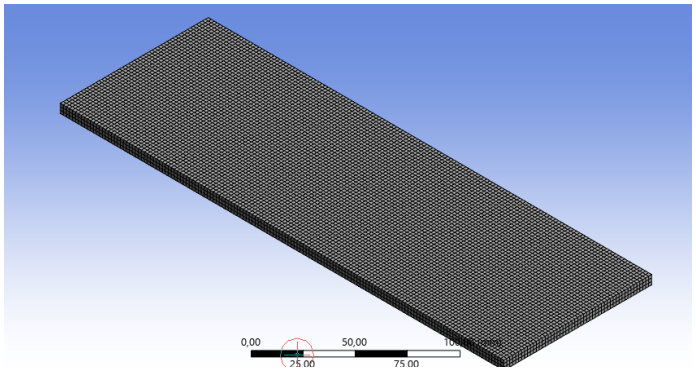


Fig. 4 ANSYS mesh with 2 mm element size (three elements through the thickness).

B. Mesh Convergence

The modal analysis was repeated with element sizes of 10, 5, 3 and 2 mm (Table 4, Fig. 5). Refining from 3 to 2 mm changes the first frequency by only 0.02%, and all six frequencies change by less than 0.03%. The mesh is therefore converged. The 3 mm mesh was used for the harmonic analysis.

Table 4 ANSYS natural frequencies (Hz) for different element sizes

Element size	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6
10 mm	46.885	305.10	333.81	715.86	876.71	1049.2
5 mm	46.782	304.39	333.38	715.66	874.63	1047.7
3 mm	46.707	303.85	332.55	715.54	872.79	1044.9
2 mm	46.696	303.77	332.47	715.47	872.55	1044.7

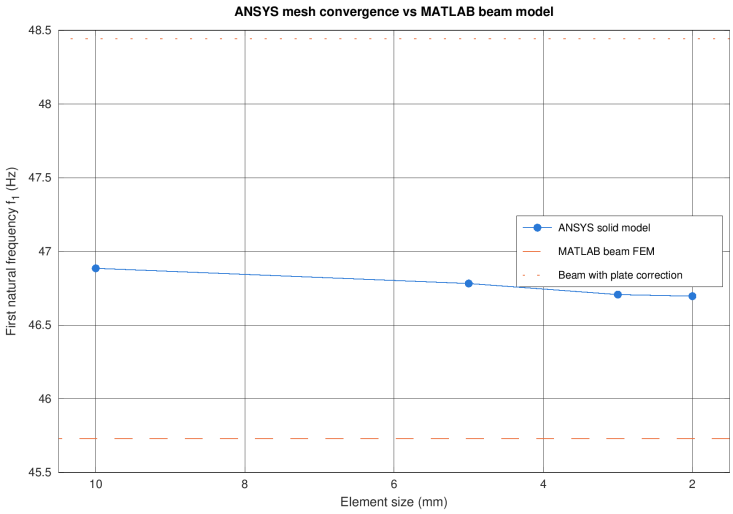


Fig. 5 Convergence of the ANSYS first natural frequency compared with the MATLAB beam model and the beam model with plate correction.

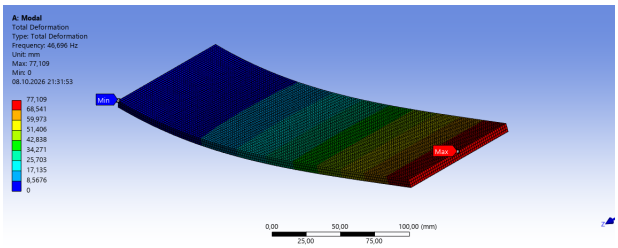
C. Mode Shapes and Comparison with MATLAB

The first four mode shapes are shown in Fig. 6. Mode 1 is the first bending mode in the thickness direction and is the mode that matters for the requirement. Mode 2 is the second bending mode, mode 3 is the first torsion mode and mode 4 is bending in the plane of the panel. The beam model cannot predict torsion and in-plane bending, so only the bending modes can be compared.

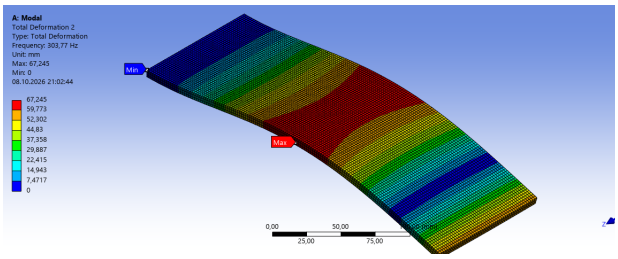
Table 5 Bending frequencies: ANSYS (2 mm mesh) versus MATLAB beam model

Mode	ANSYS, Hz	Beam FEM, Hz	Beam with plate correction, Hz	Difference ANSYS–Beam
1st bending	46.70	45.73	48.44	+2.1%
2nd bending	303.77	298.78	316.51	+1.7%
3rd bending	872.55	857.70	908.60	+1.7%

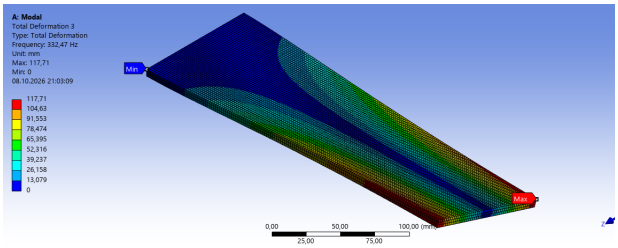
As seen in Table 5, ANSYS gives frequencies about 2% higher than the beam model. The reason is that the panel is a wide plate and not a narrow beam. In a beam, the cross-section can freely contract and expand sideways (Poisson effect), while in a wide plate this is partly prevented, which makes the plate stiffer. In the limit of a very wide plate, E should be replaced by $E/(1-\nu^2)$, which raises the frequency by $1/\sqrt{1-\nu^2} = 5.9\%$. The ANSYS results lie between the beam value and this upper limit, as expected for a plate with $b/L = 1/3$.



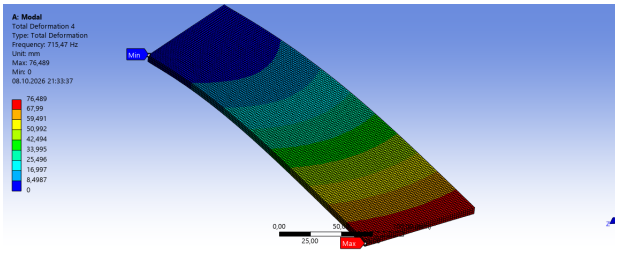
(a) Mode 1: first bending, 46.70 Hz



(b) Mode 2: second bending, 303.77 Hz



(c) Mode 3: first torsion, 332.47 Hz



(d) Mode 4: in-plane bending, 715.47 Hz

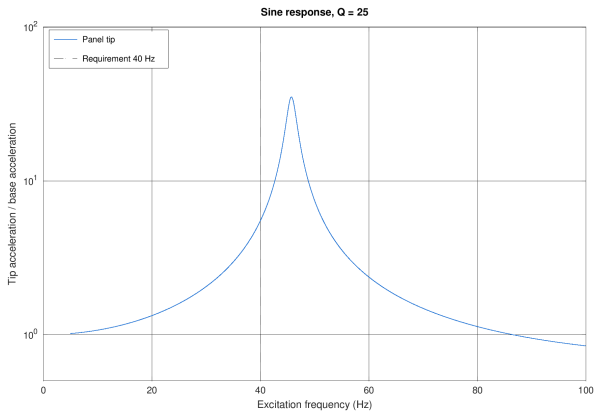
Fig. 6 First four mode shapes from ANSYS (2 mm mesh).

So the main result is that the ANSYS first frequency of 46.7 Hz meets the 40 Hz requirement with a margin of 16.7%.

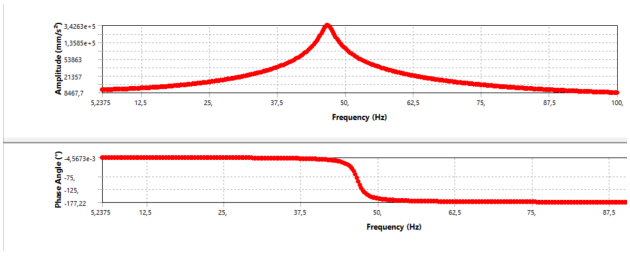
VII. Sine Vibration Response

The panel response to sine base excitation in the thickness direction was computed from 5 to 100 Hz with 2% damping. The exact sine levels depend on the launch vehicle and are given in the user's guide [1]. Since the system is linear, the analysis was done for a unit input of 1 g, and the results can simply be multiplied by the real level.

In MATLAB, modal superposition with the first four modes was used. The first mode carries 63% of the moving mass as effective modal mass, and the first four modes together carry 92%. In ANSYS, a Harmonic Response analysis (mode superposition) linked to the modal analysis was used, with the acceleration applied as base excitation at the fixed support, 400 frequency steps and absolute results.



(a) MATLAB: tip transmissibility



(b) ANSYS: tip acceleration amplitude (mm/s^2) and phase

Fig. 7 Tip response to 1 g sine base excitation.

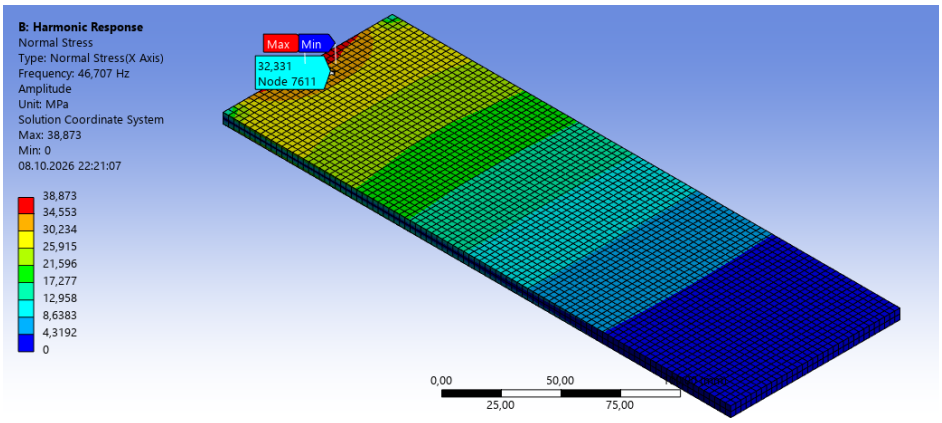


Fig. 8 ANSYS normal stress along the panel length at resonance (46.7 Hz, 1 g input). The probe value of 32.3 MPa is taken at mid-width close to the root.

Table 6 Sine response at resonance for 1 g base input

Quantity	MATLAB	ANSYS	Difference
Resonance frequency, Hz	45.7	46.7	+2.1%
Peak tip acceleration, g	35.1	34.9	−0.5%
Root bending stress, MPa	32.2	32.3	+0.4%

The results are given in Table 6. The two models agree within 1% in amplitude and stress. At resonance the tip moves about 35 times more strongly than the base. This is larger than $Q = 25$ because the tip of a cantilever moves more than the average point of the panel. The phase of the tip response changes from 0° to -180° and passes -90° at resonance. This is typical when one mode dominates the response. Above about 85 Hz the tip acceleration becomes smaller than the input, so the panel isolates higher frequencies.

The maximum stress in ANSYS (38.9 MPa) appears at the corners of the fixed face (Fig. 8). This is a stress concentration caused by the perfectly rigid boundary condition and depends on the mesh, so the probe value at mid-width was used for comparison. With a factor of safety of 1.25 on yield (a typical value; the actual factors are set by the launch provider), the margin of safety for 1 g input is

$$MS = \frac{\sigma_y}{FS \cdot \sigma} - 1 = \frac{276}{1.25 \times 32.3} - 1 = 5.8 \tag{8}$$

This means that the panel can withstand a sine input of about 6.8 g at resonance before reaching the allowable stress.

VIII. Discussion and Limitations

The three methods (hand formula, MATLAB beam model and ANSYS solid model) give a similar results, and the remaining differences have a physical explanation. Still, the work has some limitations:

- The hinge was modeled as rigid. A real hinge and its bracket add flexibility and lower the first frequency, possibly by more than the 16.7% margin. A test or a detailed hinge model is needed.
- Damping of 2% was assumed. Real damping of bolted aluminum structures is often between 1% and 5%, so the peak response may differ.
- The sine levels were normalized to 1 g. The real levels from the user's guide must be applied before a final strength check.
- The sandwich panel results come only from a simplified beam model without core shear deformation, and were not verified in ANSYS.
- Only the deployed cantilever configuration was studied. A real panel is stowed and held at several hold-down points during launch.

IX. Conclusions

A deployable solar panel of a small satellite was sized against the 40 Hz minimum frequency requirement of the SpaceX Rideshare Payload User's Guide. The main conclusions are:

1. A self-written MATLAB beam finite element code reproduced the exact cantilever frequencies with less than 0.001% error and agreed with the Rayleigh estimate within 1%.
2. A solid aluminum panel needs a thickness of 6 mm to pass the requirement with margin, which leads to a heavy panel (486 g).
3. An aluminum honeycomb sandwich panel with an 8 mm core reaches 55.3 Hz with only 65 g, an 87% mass reduction. This explains why real solar panels use sandwich construction.
4. The ANSYS solid model of the 6 mm aluminum panel gives $f_1 = 46.7$ Hz, a 16.7% margin over the requirement, with a converged mesh. It differs from the beam model by only 2%, and the difference is explained by plate effects.
5. Under 1 g sine base excitation, the peak tip acceleration is about 35 g and the root stress is 32.3 MPa. MATLAB and ANSYS agree within 1%.

Future work should include an ANSYS model of the sandwich panel with core shear, a model of the hinge stiffness, the real sine and random vibration levels from the user's guide and, if possible, a simple vibration test to measure the first frequency and damping.

Acknowledgments

The finite element analysis was carried out with ANSYS Mechanical 2026 R1 Student, the geometry was created in SolidWorks, and the beam model was implemented in MATLAB.

References

[1] SpaceX, *Rideshare Payload User's Guide*, Space Exploration Technologies Corp., Hawthorne, CA, 2026.

[2] ASM International, *ASM Handbook, Vol. 2: Properties and Selection: Nonferrous Alloys and Special-Purpose Materials*, ASM International, Materials Park, OH, 1990.

[3] Blevins, R. D., *Formulas for Natural Frequency and Mode Shape*, Van Nostrand Reinhold, New York, 1979.

[4] Rao, S. S., *Mechanical Vibrations*, 6th ed., Pearson, Hoboken, NJ, 2017.

[5] Logan, D. L., *A First Course in the Finite Element Method*, 6th ed., Cengage Learning, Boston, MA, 2017.

[6] European Cooperation for Space Standardization, *Spacecraft Mechanical Loads Analysis Handbook*, ECSS-E-HB-32-26A, ESA Requirements and Standards Division, Noordwijk, The Netherlands, 2013.

[7] Zenkert, D., *An Introduction to Sandwich Construction*, Engineering Materials Advisory Services, London, 1995.

[8] Wijker, J. J., *Spacecraft Structures*, Springer, Berlin, 2008.